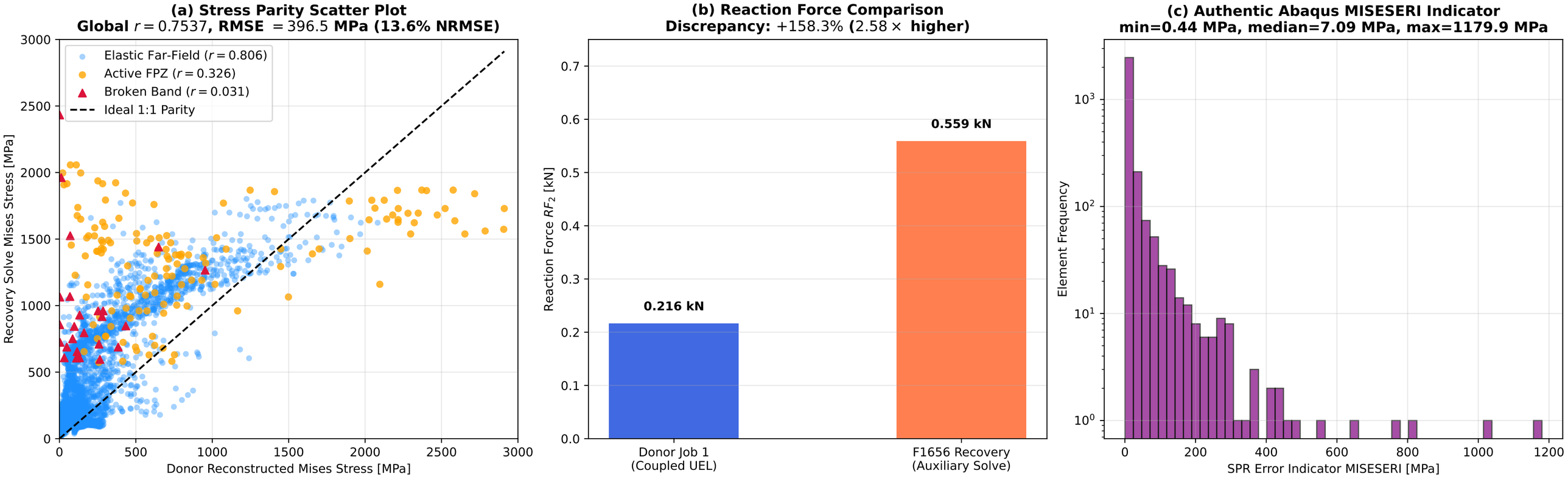

(a) Stress Parity Scatter Plot
Global r = 0.7537, RMSE = 396.5 MPa (13.6% NRMSE)
Elastic Far-Field (r = 0.806)
Active FPZ (r = 0.326)
Broken Band (r = 0.031)
Ideal 1:1 Parity
Recovery Solve Mises Stress [MPa]
Donor Reconstructed Mises Stress [MPa]
0
500
1000
1500
2000
2500
3000
(b) Reaction Force Comparison
Discrepancy: +158.3% (2.58 × higher)
Reaction Force RF_2 [kN]
0.216 kN
0.559 kN
Donor Job 1
(Coupled UEL)
F1656 Recovery
(Auxiliary Solve)
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
(c) Authentic Abaqus MISESERI Indicator
min=0.44 MPa, median=7.09 MPa, max=1179.9 MPa
Element Frequency
SPR Error Indicator MISESERI [MPa]
10^0
10^1
10^2
10^3
0
200
400
600
800
1000
1200